



Vrije Universiteit Amsterdam

Faculty of Humanities

Master in Linguistics; Text Mining

Applied Text Mining

Assignment 1: Scope Annotation and Error Analysis

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Negation scope detection is an NLP task that helps determine which tokens within a sentence are affected by a negation cue. In the 'file-17-to-annotate.csv' there are 11 unique negation forms and a total of 32 negation cue instances. After annotating the scope and event of these negation cues, the most challenging cases in scope detection are highlighted.

1. *Gold annotation:* Our client smoothed down [his] **un**[brushed hair] and felt [his] **un**[shaven chin].

Annotation: Our client smoothed down his **un**[brushed hair] and felt [his] **un**[shaven chin].

This sentence presents with two negations, both in the form of an adjective prefix. In both instances annotated by me, the pronoun 'his' was disregarded. In my opinion, the pronoun 'his' was not directly affected by the negation cue, however, in the gold labels it is marked as in scope. Upon reflection, this discrepancy might stem from my interpretation of the guidelines, where I considered only the negated attributes ('brushed hair' and 'shaven chin') as relevant.

2. *Gold annotation:* [He is] **not** [particularly intelligent] – **not** [a man likely to be congenial to a quick-witted Latin].

Annotation: [He is] **not** [particularly intelligent] – **not** [a man likely to be congenial to a quick-witted Latin].

This sentence also contains two negation cue instances. In my annotations, the scope of the second cue was only 'a man likely to be congenial to a quick-witted Latin', while in the original dataset it also contained 'He is'. This likely stems from the fact that the two parts of this sentence were separated by an n-dash, which led me to believe that they would be treated as separate. This highlights the importance of understanding how punctuation marks and clause boundaries affect the scope.

3. *Gold annotation:* [I did]**n't** [say so], Mr. Holmes; [I did]**n't** [say so].

Annotation: [I did]**n't** [say so], [Mr. Holmes]; [I did]**n't** [say so].

In this case, my disagreement with the gold labels was that ‘Mr. Holmes’ was also in the scope of the first negation cue. Given the conversational context, I believed that Mr. Holmes was affected directly by it.

4. *Gold annotation:* [The TRUE voodoo-worshipper attempts] **nothing** [of importance **without** [certain sacrifices] which are intended to propitiate [his] **un**[clean gods]].

Annotation: [The TRUE voodoo-worshipper attempts] **nothing** [of importance **without** [certain sacrifices] which are intended to propitiate his **un**[clean gods]].

In this sentence three different negation cues are present, and their scope is tangled. In my annotations, I did not include ‘his’ in the scope of the last negation, similarly to the first sentence. This repeated disagreement shows a pattern in my annotations, as I consistently interpreted possessive pronouns as outside the scope of negation.